

XHDPF: Institutional Transferability Protocol

A 4-Phase Adaptation Framework for Cross-University Deployment

Phase 1: Governance & Canonical Schema Mapping

Ingest local administrative (SIS) and LMS databases into the canonical DAMA-DMBOK schema. Enforce SHA-256 cryptographic pseudonymization on student IDs.

Phase 2: Domain Adaptation & Recalibration

Apply domain adaptation (Maximum Mean Discrepancy alignment) and fine-tune the CTGAN generator and Late Fusion weights (α) on local institutional training folds.

Phase 3: Subgroup Fairness & XAI Validation

Execute automated fairness audits to ensure Disparate Impact ($DI \geq 0.80$) and Equalized Odds Difference ($EOD \leq 0.05$). Validate leakage-free LIME/TreeSHAP local attributions.

Phase 4: Advisor Cognitive Load & Deployment

Integrate interpretable intervention alerts into academic advising workflows. Assess advisor cognitive load using System Usability Scale ($SUS \geq 80$) and NASA-TLX instruments.